

## LAB - SWR302

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### 1. Use Cases

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|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------|
| UC ID and Name:    | UC57- Create Purchase Proposal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |              |
| Created By:        | QuynhNTT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Date Created:     | June 3, 2026 |
| Primary Actor:     | Sales Staff                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Secondary Actors: | NONE         |
| Trigger:           | Sales Staff selects to create a new purchase proposal from the proposal management screen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |              |
| Description:       | This use case allows the Sales Staff to create a new purchase proposal for buying devices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |              |
| Preconditions:     | <ul style="list-style-type: none"> <li>- The Sales Staff has successfully logged into the system.</li> <li>- The Sales Staff has permission to create purchase proposals.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                       |                   |              |
| Postconditions:    | - A new purchase proposal is created with "Pending" status, waiting for Sales Manager approval.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |              |
| Normal Flow:       | <ol style="list-style-type: none"> <li>1. Sales Staff navigates to the purchase proposal management feature.</li> <li>2. Sales Staff requests to create a new purchase proposal.</li> <li>3. System presents a form for entering proposal details (items, quantities, estimated prices, supplier, notes...).</li> <li>4. Sales Staff fills in the proposal information.</li> <li>5. Sales Staff submits the proposal.</li> <li>6. System validates that all required fields are provided.</li> <li>7. System saves the proposal and displays a success message.</li> </ol> |                   |              |
| Alternative Flows: | At step 3, instead of manually entering items, <ol style="list-style-type: none"> <li>1. Sales Staff chooses to import from a Excel file.</li> <li>2. System parses the file and pre-fills the item list.</li> <li>3. Sales Staff reviews and adjusts the imported data.</li> <li>4. Return to step 5 of normal flow.</li> </ol>                                                                                                                                                                                                                                           |                   |              |
| Exceptions:        | E1: At step 6, if required fields are missing, system prevents submission and displays an error indicating which fields are missing.<br>E2: At step 6, if staff enters quantity smaller than 0, system prevents submission and requests to enter again.                                                                                                                                                                                                                                                                                                                    |                   |              |
| Priority:          | 2 - High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |              |
| Frequency of Use:  | 2 -Weekly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |              |
| NFR:               | <ul style="list-style-type: none"> <li>- Security: Only accounts with the purchase_proposal.create privilege can execute this request. Any unauthorized attempt must be rejected immediately without exposing the creation interface.</li> <li>- Performance: The proposal creation and validation must complete within less than 3 seconds.</li> </ul>                                                                                                                                                                                                                    |                   |              |
| Business Rules:    | At step 4: Each proposal must contain at least one item with a specified quantity and estimated price.<br>At step 4: The device quantity must be greater than zero.                                                                                                                                                                                                                                                                                                                                                                                                        |                   |              |
| Other Information: | This use case is the starting point of the procurement process. All purchase orders are derived from approved proposals.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |              |

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| Assumptions: | It is assumed that the Sales Staff has the necessary information about required items, including estimated prices from suppliers. |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|

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|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------|
| UC ID and Name:    | UC-58- Approve Purchase Proposal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |              |
| Created By:        | QuynhNTT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Date Created:     | June 3, 2026 |
| Primary Actor:     | Sales Manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Secondary Actors: | NONE         |
| Trigger:           | Sales Manager selects a pending purchase proposal from the approval list to review and approve.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |              |
| Description:       | This use case allows the Sales Manager to review and approve purchase proposals submitted by Sales Staff.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |              |
| Preconditions:     | <ul style="list-style-type: none"> <li>- The Sales Manager has successfully logged into the system.</li> <li>- The Sales Manager has permission to approve purchase proposals.</li> <li>- At least one purchase proposal with "Pending" status exists.</li> </ul>                                                                                                                                                                                                                                                                                                           |                   |              |
| Postconditions:    | - The purchase proposal status is updated to "Approved".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |              |
| Normal Flow:       | <ol style="list-style-type: none"> <li>1. Sales Manager navigates to the purchase proposal queue.</li> <li>2. System displays a list of proposals with "Pending" status.</li> <li>3. Sales Manager selects a proposal to review.</li> <li>4. System displays the full proposal details including items, quantities, prices, and total...</li> <li>5. Sales Manager reviews the proposal information.</li> <li>6. Sales Manager approves the proposal.</li> <li>7. System updates the proposal status to "Approved" and notifies the Sales Staff of the approval.</li> </ol> |                   |              |
| Alternative Flows: | <p>At step 2, Sales Manager selects multiple proposals instead of one.</p> <ol style="list-style-type: none"> <li>1. System displays a summary list of selected proposals.</li> <li>2. Sales Manager reviews the summaries.</li> <li>3. Sales Manager approves all selected proposals at once.</li> <li>4. System updates all selected proposals to "Approved".</li> </ol>                                                                                                                                                                                                  |                   |              |
| Exceptions:        | <p>At step 3,</p> <ul style="list-style-type: none"> <li>- The system detects that the proposal has been pending for more than 30 days and the prices may no longer be valid.</li> <li>- The system displays a warning asking the Manager to request the Sales Staff to update the proposal with current prices before proceeding.</li> </ul>                                                                                                                                                                                                                               |                   |              |
| Priority:          | 2 - High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |              |
| Frequency of Use:  | 2 - Weekly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |              |
| NFR:               | <ul style="list-style-type: none"> <li>- Security: Only accounts with the purchase_proposal.approve privilege can execute this request. Any unauthorized attempt must be rejected immediately.</li> <li>- Performance: The approval action and status update must complete within less than 2 seconds.</li> </ul>                                                                                                                                                                                                                                                           |                   |              |
| Business Rules:    | At step 4: The estimated unit price of each item must not exceed 120% of the latest market reference price stored in the system.                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |              |
| Other Information: | This use case is a critical control point in the procurement process, ensuring that all purchases are properly reviewed before proceeding.                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |              |
| Assumptions:       | It is assumed that the Sales Manager has sufficient knowledge to evaluate the necessity and reasonableness of the proposed purchases.                                                                                                                                                                                                                                                                                                                                                                                                                                       |                   |              |

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|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------|
| UC ID and Name:    | UC-60- Create Purchase Order                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |              |
| Created By:        | QuynhNTT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Date Created:     | June 3, 2026 |
| Primary Actor:     | Sales Manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Secondary Actors: | None         |
| Trigger:           | Sales Manager navigates to the purchase order management feature and requests to create a new consolidated monthly purchase order from approved proposals.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |              |
| Description:       | This use case allows the Sales Manager to aggregate all approved purchase proposals into a consolidated purchase order and submit for final approval.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |              |
| Preconditions:     | <ul style="list-style-type: none"> <li>- The Sales Manager has successfully logged into the system.</li> <li>- The Sales Manager has permission to create purchase orders.</li> <li>- At least one purchase proposal with "Approved" status exists.</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |                   |              |
| Postconditions:    | - A new purchase order is created with "Pending" status, waiting for CEO approval.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |              |
| Normal Flow:       | <ol style="list-style-type: none"> <li>1. Sales Manager navigates to the purchase order management feature.</li> <li>2. Sales Manager requests to create a monthly purchase order.</li> <li>3. System automatically retrieves all approved purchase proposals from the current month.</li> <li>4. System aggregates the items from approved proposals into a purchase order.</li> <li>5. System displays the consolidated purchase order for review.</li> <li>6. Sales Manager reviews the purchase order details and submits.</li> <li>7. System saves the purchase order and displays a success message.</li> </ol> |                   |              |
| Alternative Flows: | At step 3, instead of auto-retrieving all approved proposals, <ol style="list-style-type: none"> <li>1. Sales Manager chooses to manually select specific proposals.</li> <li>2. System displays a list of all approved proposals.</li> <li>3. Sales Manager selects the desired proposals.</li> <li>4. System aggregates only the selected proposals into the purchase order.</li> <li>5. Return to step 5 of normal flow.</li> </ol>                                                                                                                                                                                |                   |              |
| Exceptions:        | At step 3, if there are no proposals with "Approved" status — System displays a message indicating no proposals to collect and terminates the creation process.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                   |              |
| Priority:          | 2 - High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |              |
| Frequency of Use:  | 3 - Monthly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |              |
| NFR:               | <ul style="list-style-type: none"> <li>- Security: Only accounts with the purchase_order.create privilege can create this request. Only accounts with the purchase_order.approve privilege can approve it. Any unauthorized attempt must be rejected immediately.</li> <li>- Performance: The aggregation logic and order creation must complete within less than 5 seconds.</li> </ul>                                                                                                                                                                                                                               |                   |              |
| Business Rules:    | At step 4: Only proposals with status "Approved" are available for collecting into a purchase order.<br>At step 4: Each approved proposal can only be included in one purchase order to prevent duplicate procurement.                                                                                                                                                                                                                                                                                                                                                                                                |                   |              |
| Other Information: | This use case represents the core step in the procurement planning phase before it proceeds to CEO approval before procurement execution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |              |
| Assumptions:       | It is assumed that the Sales Manager has reviewed all approved proposals before consolidation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |              |

### Priority

| Value | Meaning                                        |
|-------|------------------------------------------------|
| 1     | Critical — system cannot function without this |
| 2     | High — important for core business process     |
| 3     | Medium — enhances user experience              |

|   |                          |
|---|--------------------------|
| 4 | Low — future enhancement |
|---|--------------------------|

### Frequency of Use

| Value | Meaning   |
|-------|-----------|
| 1     | Daily     |
| 2     | Weekly    |
| 3     | Monthly   |
| 4     | quarterly |

## 2. Business Rules

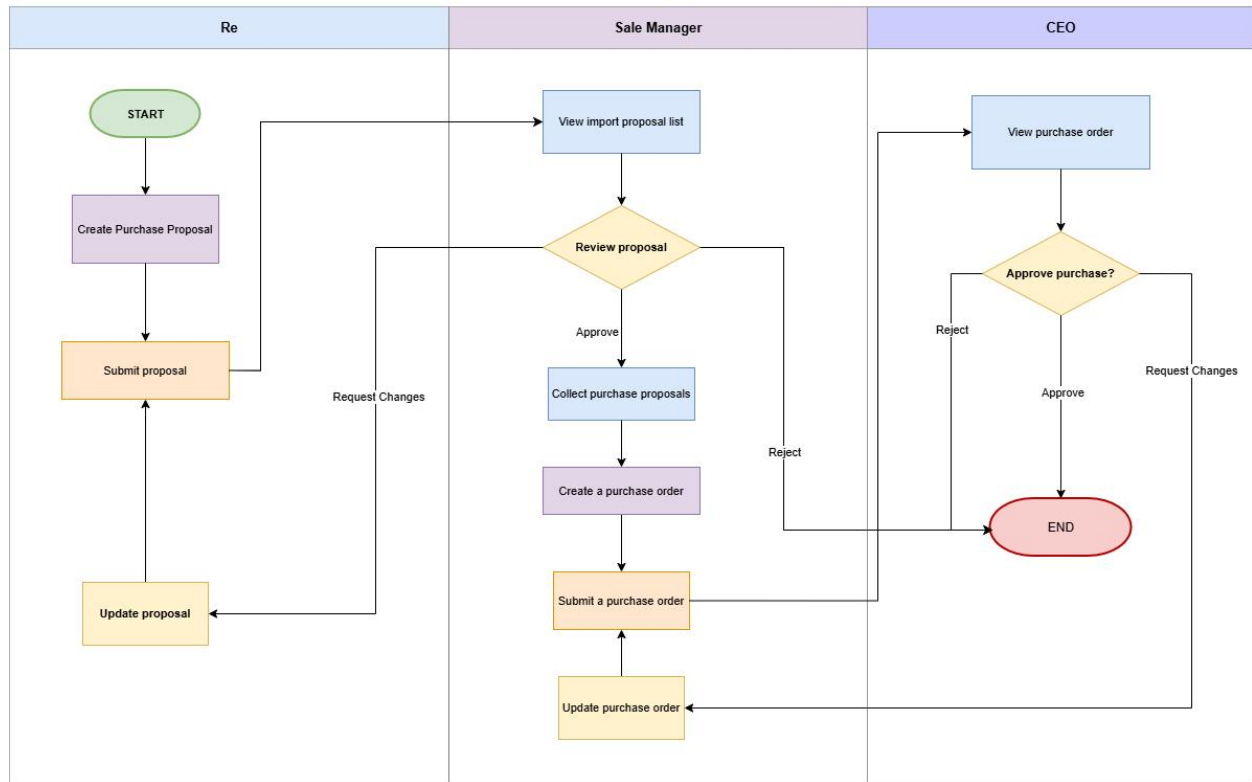
| Type           | Description                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------|
| Constraint     | The system must not allow the deactivation of any user account that holds the "Admin" role                            |
| Action enabler | On the last day of a calendar quarter, generate the inventory check reports for import/export history in that quarter |

## 3. Non functional Requirements

| Group       | Description                                                                                                                                                 | Evaluation Criteria & Measurement                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Security    | The system shall enforce RBAC (Role-Based Access Control). Only users with appropriate permissions shall be allowed to perform operations.                  | Every request from a non-privileged account must be blocked and return HTTP 403.  |
| Performance | The system shall process an import receipt approval containing up to 300 serial numbers within an acceptable time to avoid disrupting warehouse operations. | The approval transaction for 300 serial numbers must complete in under 5 seconds. |

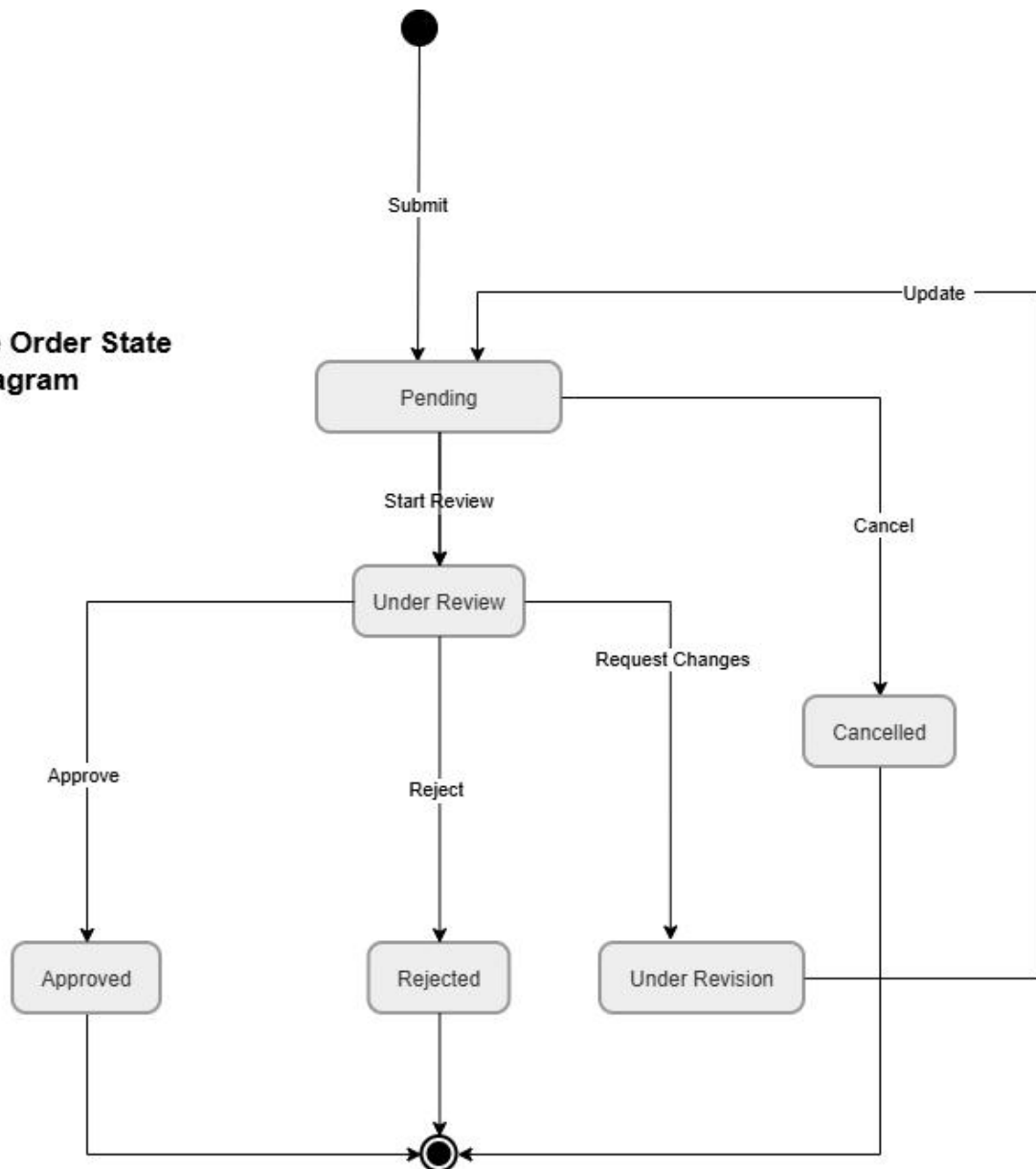
## 4. Swimlane

## Purchase order process



## 5. State Diagram

**Purchase Order State  
Diagram**



**6. ERD**

